

INCOSE Systems Engineering Template: Simple Alarm Clock

1. Business or Mission Analysis

1.1 Problem Statement

Users need a reliable and simple way to keep track of the current time and be awakened at a specific pre-set time. Traditional methods or smart devices might be overly complex or prone to failure if power is lost or they require extensive configuration. The proposed simple alarm clock provides a dedicated, easy-to-use device for these essential functions.

1.2 Mission Objectives

- To display the current time clearly and accurately.
- To allow the user to easily set an alarm time.
- To sound an audible alarm when the current time matches the set alarm time.
- To provide a simple means to disable the alarm.

1.3 Key Stakeholders (Business Level)

SysMLv2 Mapping: part def with actor or stakeholder applied.

- **User / Sleeper:** The primary person using the alarm clock to wake up on time.
- **Manufacturer:** The organization building and selling the alarm clock.

2. Stakeholder Needs and Requirements Definition

2.1 Use Cases / Operational Scenarios

SysMLv2 Mapping: use case def.

- **View Time:** The User observes the current time displayed on the clock.
- **Set Alarm:** The User interacts with the clock's interface to specify the desired wake-up time.
- **Wake Up / Sound Alarm:** The system automatically sounds an alert when the alarm time is reached.
- **Dismiss Alarm:** The User turns off the sounding alarm.

2.2 Measures of Effectiveness (MOEs)

SysMLv2 Mapping: attribute def or constraint blocks.

- **Time Accuracy:** The clock must not deviate more than 1 minute per month.
- **Alarm Volume:** The alarm must be loud enough to wake a sleeping user (e.g., > 70 dB).
- **Usability:** The process of setting an alarm should take less than 15 seconds for an average user.

3. System Requirements Definition

3.1 Functional Requirements

SysMLv2 Mapping: requirement def.

- **REQ-FUNC-01 (Display Time):** The system shall continuously display the current time in hours and minutes.
- **REQ-FUNC-02 (Set Alarm Time):** The system shall provide a mechanism for the user to set an alarm time.
- **REQ-FUNC-03 (Trigger Alarm):** The system shall trigger an audible alarm when the current time matches the set alarm time.
- **REQ-FUNC-04 (Dismiss Alarm):** The system shall provide a mechanism to stop the audible alarm once it is triggered.

3.2 Non-Functional / Quality Requirements

SysMLv2 Mapping: requirement def.

- **REQ-PERF-01 (Volume):** The alarm volume shall be at least 70 dB measured at a distance of 1 meter.
- **REQ-PERF-02 (Accuracy):** The internal clock shall maintain time with an accuracy of +/- 1 minute per month.
- **REQ-ELEC-01 (Power Source):** The system shall operate on standard household AC power with an optional battery backup.

3.3 System Constraints

SysMLv2 Mapping: requirement def or constraint def.

- **CON-01 (Size):** The system footprint shall not exceed 15 cm x 15 cm to fit on a standard nightstand.

4. Architecture Definition

4.1 Logical Architecture / Subsystems

SysMLv2 Mapping: part def.

- **Logical System:** LogicalAlarmClock
- **Subsystem 1:** Timekeeper - Maintains current time and alarm time.
- **Subsystem 2:** UserInterface - Handles inputs (setting time/alarm, dismissing alarm) and outputs (displaying time).
- **Subsystem 3:** AlarmTrigger - Compares current time to alarm time and controls the sound emission.

4.2 Interfaces and Interactions

SysMLv2 Mapping: interface def, port def, connection.

- UserInterface sends 'SetAlarmTime' data to Timekeeper.
- Timekeeper continuously provides 'CurrentTime' and 'AlarmTime' to AlarmTrigger.
- AlarmTrigger sends 'SoundSignal' to the UserInterface (specifically, its sound output component).

4.3 System Behavior

SysMLv2 Mapping: action def, state def.

- **State: NormalOperation:** Clock is displaying time.
- **State: AlarmSounding:** Alarm has been triggered and is sounding.
- **Transition:** NormalOperation -> AlarmSounding (triggered when CurrentTime == AlarmTime).
- **Transition:** AlarmSounding -> NormalOperation (triggered by User dismissing alarm).

5. Design Definition

5.1 Physical Components

SysMLv2 Mapping: part def (often inheriting from logical parts).

- **Physical System:** PhysicalAlarmClock
- **Component 1:** Microcontroller (Realizes Timekeeper and AlarmTrigger logic)
- **Component 2:** LED_Display (Realizes visual part of UserInterface)
- **Component 3:** Buzzer (Realizes audio part of UserInterface/AlarmTrigger)
- **Component 4:** Buttons (Realizes input part of UserInterface)
- **Component 5:** PowerSupply (AC adapter and battery backup)

5.2 Allocation Matrix

SysMLv2 Mapping: allocation relationships or allocate dependencies.

- PhysicalAlarmClock -> LogicalAlarmClock
- Microcontroller -> Timekeeper
- Microcontroller -> AlarmTrigger
- LED_Display -> UserInterface (visual output)
- Buzzer -> UserInterface (audio output)
- Buttons -> UserInterface (inputs)

LLM Instruction: When processing this file, ensure that every requirement listed in Section 3 has a corresponding satisfy relationship from an architectural element in Section 4 or 5.